SPECTRA

Band-Routed Embedding and Stage-Wise LoRA for **Cross-Sensor Fine-Tuning of Geospatial Foundation Models**

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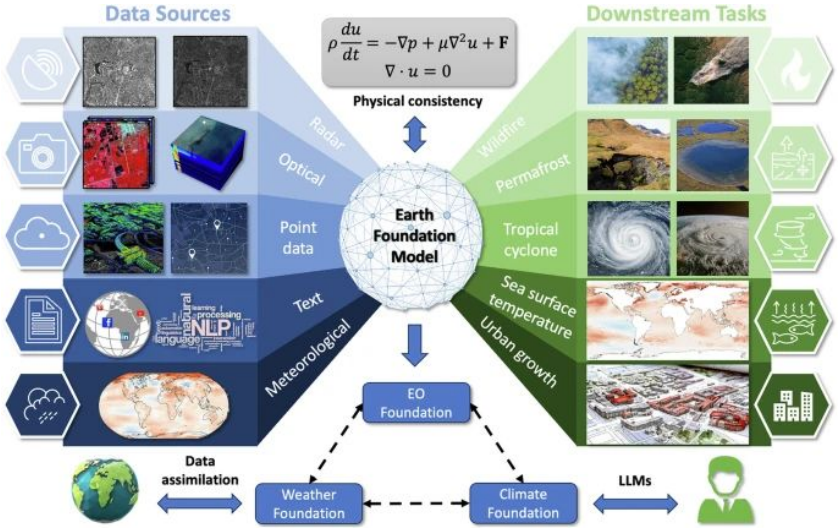
Paper

Code

Website and CV

Geospatial foundation models

GeoFMs have shown success in Earth observation tasks by leveraging self-supervised learning on vast amounts of unlabeled satellite data.



Pretraining

Learn from diverse, unlabeled observations.

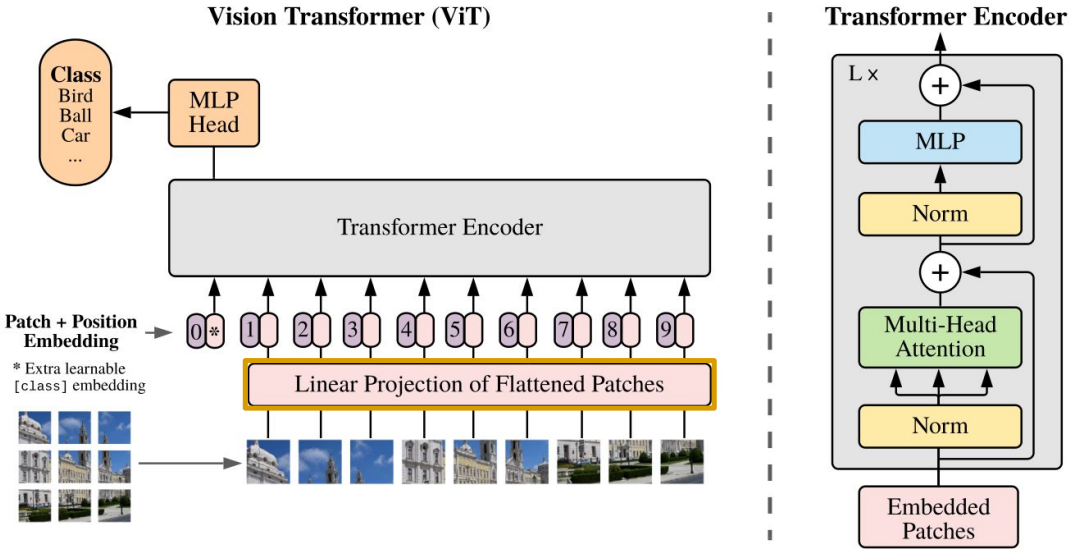
Fine-tuning

Adapt the pretrained model to a task using labeled data.

Zhu et al., On the foundations of Earth foundation models, Commun. Earth Environ. 7, 103 (2026)

Vision Transformer (ViT) and patch embedding

Many geospatial foundation models use ViT-based encoders.



Patch embedding

- A **learned linear projection** turns each patch into a fixed-length feature vector, or **token**.
- Each patch includes all input channels (RGB in the original ViT).

The pretrained projection expects a specific number and order of input channels.

Original ViT classification architecture · Dosovitskiy et al., ICLR 2021, Fig. 1.

An example of spectral mismatch: what reaches the model?

Sentinel-2 flood dataset, measured in **13 spectral bands**.

Visible light (RGB)
B04 / B03 / B02



Visible appearance

Near-infrared (NIR)
B08 · ≈ 0.84 μm



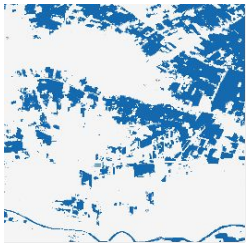
Water–vegetation contrast

Shortwave infrared (SWIR)
B11 · ≈ 1.61 μm



Water / moisture contrast

Ground truth



NIR and SWIR contain signals that RGB alone cannot show.

An example of spectral mismatch: what reaches the model?

Sentinel-2 flood dataset, measured in **13 spectral bands**.

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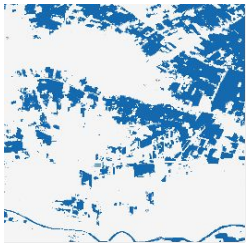
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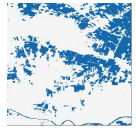


Select RGB B04 / B03 / B02

Pre-trained ScaleMAE encoder

Patch embedding → One RGB patch → one token → **Transformer Encoder**

Decoder



Predicted Flood mask

The other 10 bands are unused—including NIR and SWIR.

Two questions when adapting a GeoFM

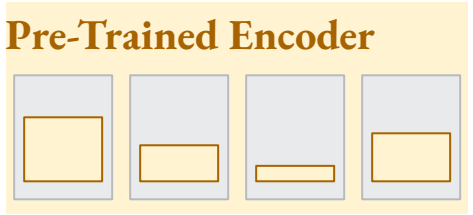
1 How do we use all the bands?

- Band selection discards information.
- Keep the expected input bands while bringing in useful information from the other bands.



2 Where should the model adapt?

- Learn small weight corrections and choose how much to learn .
- Reduce number of trainable parameters.



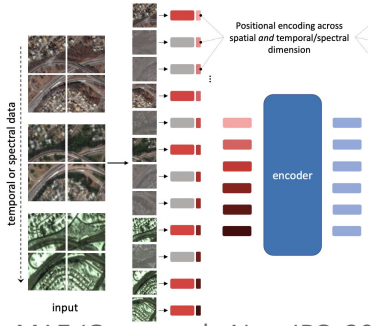
Different encoder stages, different adaptation capacity

Goal: use richer sensor information with fewer trainable parameters.

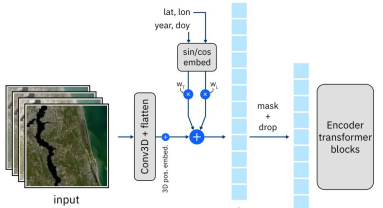
Three GeoFMs, different input interfaces

GeoFM	Input bands	Pre-training data	Original downstream tasks tested by authors
SatMAE	**10 bands (3 groups)**	Sentinel-2	BigEarthNet
Prithvi-EO-2.0	**6 bands**	Harmonized Landsat/ Sentinel-2	Sen1Floods11, HLS Burn Scars, HLS–MERRA2 GPP Flux
ScaleMAE	**3 RGB bands**	fMoW	fMoW

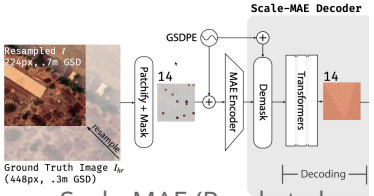
- **In original papers, the GeoFMs were tested on downstream tasks with the same input bands as pre-training data**



SatMAE (Cong et al., NeurIPS, 2022).



Prithvi-EO-2.0 (Szwarcman et al., IEEE TGRS, 2026)

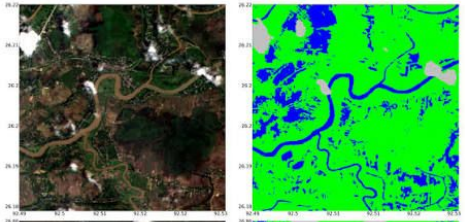


Scale-MAE (Reed et al., ICCV, 2023).

Four downstream Sentinel-2 datasets

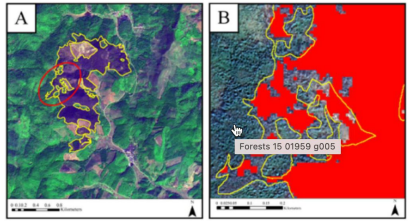
Sen1Floods11

13 bands Binary flood segmentation



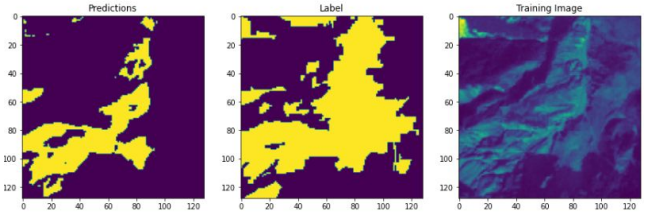
FireScars

6 bands Binary burn-scar segmentation



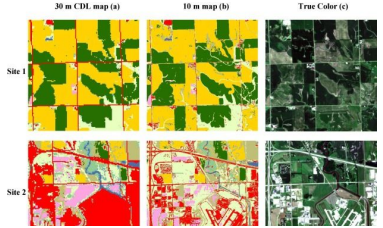
Landslide4Sense

14 bands Landslide segmentation

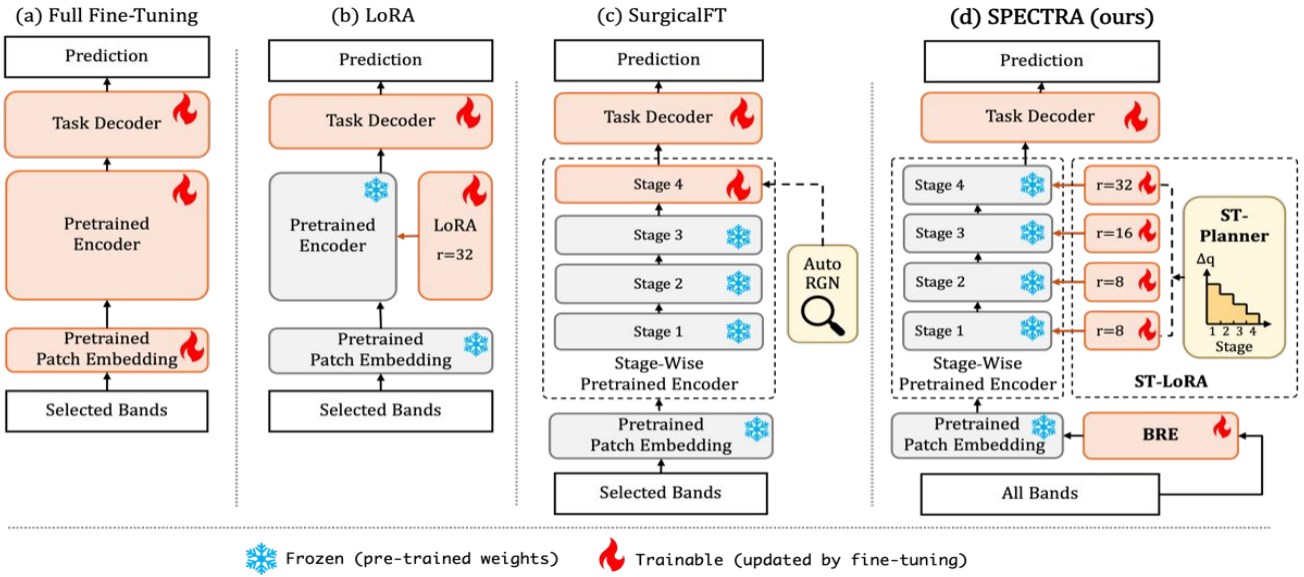


SA Crop Type

12 bands 10-class segmentation

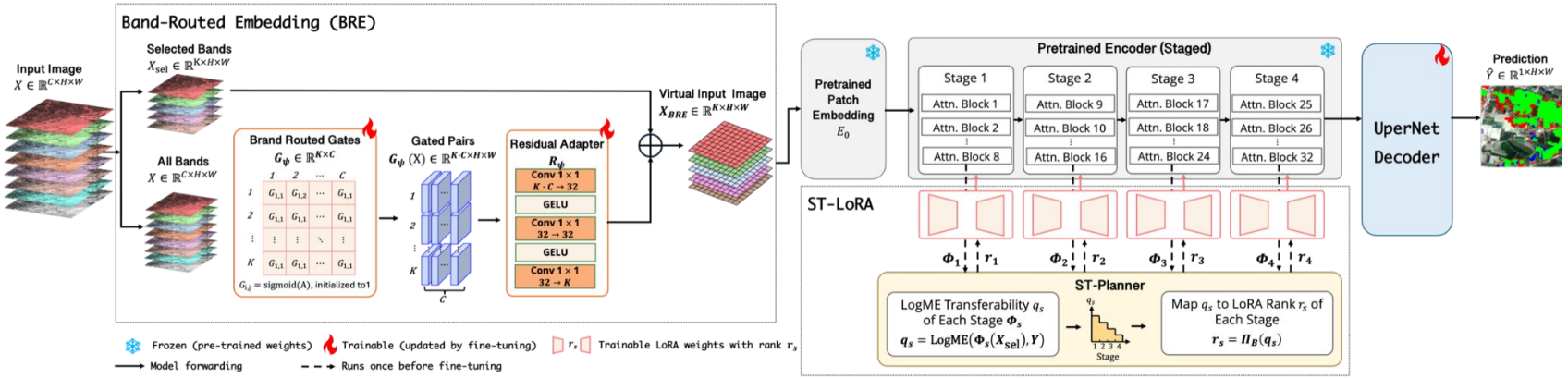


Related works: fine-tuning methods



- Band selection preserves compatibility but discards information.
- Full fine-tuning updates the entire backbone.
- LoRA assigns uniform capacity to the whole encoder.
- Surgical fine-tuning can still unfreeze a large stage.

SPECTRA separates input adaptation from encoder adaptation



SPECTRA has **BRE** and **ST-LoRA** to preserve the tokenizer and spend adaptation capacity selectively:

1. **Band-Routed Embedding (BRE)**

   C observed bands $\rightarrow$ K virtual pretrained-compatible bands

2. **Stage-wise Transferability-aware LoRA (ST-LoRA)**

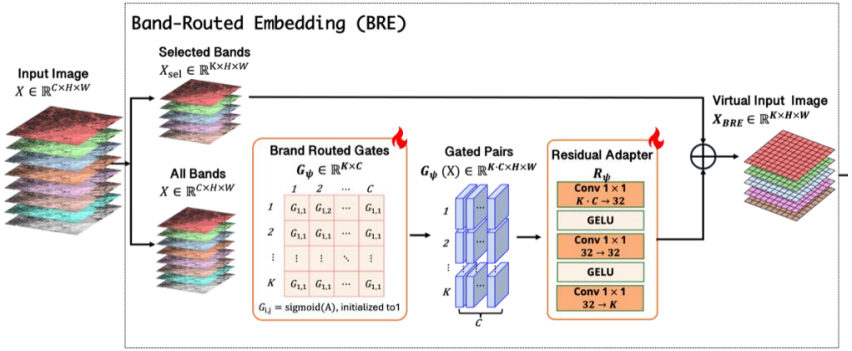
   Stage transferability $\rightarrow [r_1, r_2, \ldots, r_S]$

Band-Routed Embedding (BRE)

BRE adds an all-band residual to a physical anchor.

- **Selected-band anchor**: retain the physically matched selected bands
- **All-band residual path**: allow every observed band to influence every virtual band

$$X_{\mathrm{BRE}} = X_{\mathrm{sel}} + \Delta_\theta(X)$$



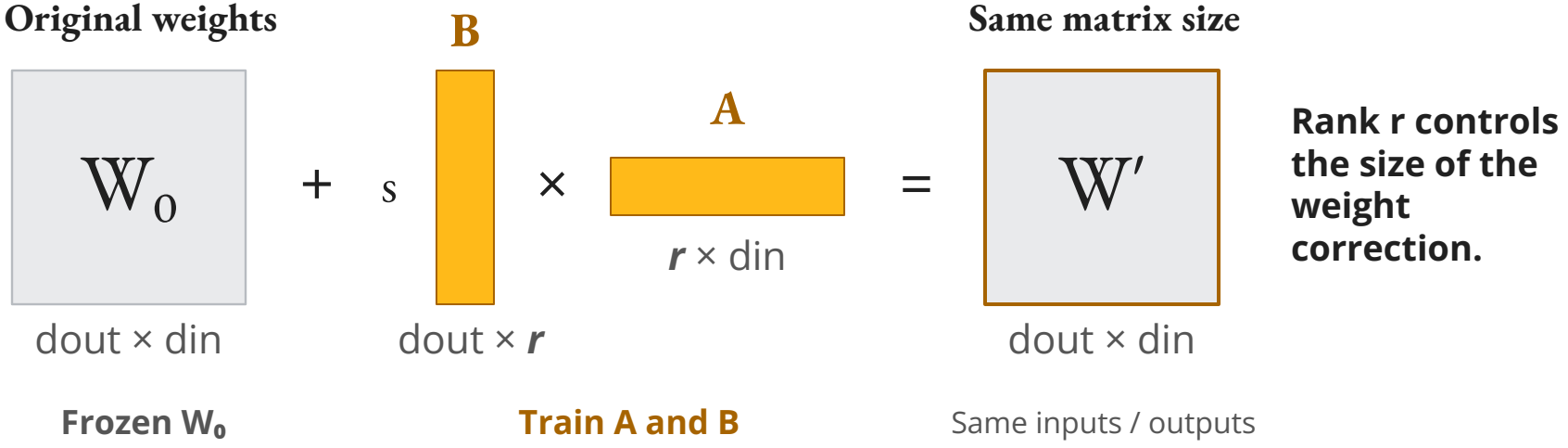
- **Stabilized initialization**: initialize the residual path to zero when training starts

$$\Delta\theta(X) = 0 \quad \Rightarrow \quad X_{\mathrm{BRE}} = X_{\mathrm{sel}}$$

Extra bands **refine, not replace**, the image that is input into the pre-trained GeoFM.

LoRA: learn a compact correction to existing weights

- LoRA freezes pre-trained weights and learns two small matrices for each attention block (a weight matrix) in the encoder.



Rank r controls the size of the weight correction.

- **Small trainable factors are added; encoder depth and width stay the same.**
- ***r:* LoRA rank**

LoRA: Hu et al. (2021). The scale factor s controls the update magnitude. Diagram shows one adapted weight matrix.

Stage-wise Transferability-aware LoRA (ST-LoRA)

ST-LoRA converts transferability into a rank schedule. It runs **one time** before fine-tuning:

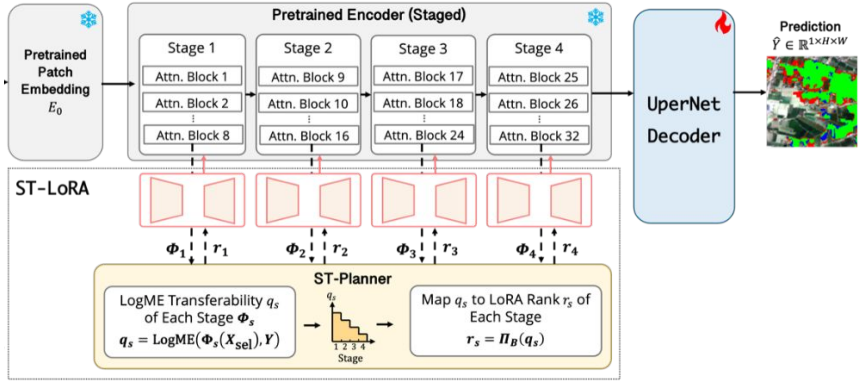
1. Divide the frozen encoder into S stages.
2. Extract frozen features from each stage Φ_s .
3. Estimate stage transferability using LogME[1]

$$q_s = \mathrm{LogME}(\Phi_s, Y)$$

4. Convert the score profile into discrete LoRA ranks.

$$r_s \in \mathcal{R}(r\mathrm{ref}), \qquad \sum s = 1^S r_s \leq S\, r_{\mathrm{ref}}$$

5. Use this stage-wise rank schedule during fine-tuning.



Stage	1	2	3	4
Standard LoRA-32	32	32	32	32
ST-LoRA	**16**	**32**	**16**	**16**

[1] Transferability metric: LogME, You et al., 2021

Experiment 1: Overall Main Comparison

Baseline methods for overall comparison of the whole fine-tuning pipeline:

Pipeline	Spectral adaptation	Encoder adaptation
Linear Probe	Band selection	Freeze the encoder
LoRA	Band selection	Train added LoRA weights
Surgical-FT	Band selection	Train a selected portion of the encoder
Full-FT	Band selection	Train the entire encoder
SPECTRA (Ours)	**BRE**	**ST-LoRA**

* All pipelines use the same decoder UperNet.

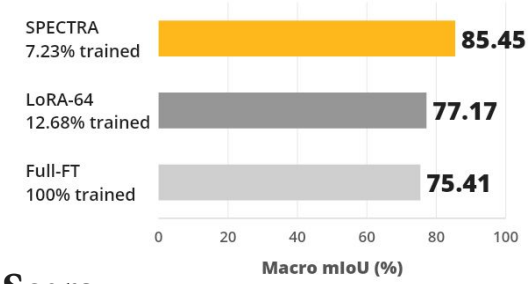
Results 1: ScaleMAE gains accuracy with fewer trainable weights

Table. Main comparison of different pipelines across all downstream datasets applied to one of the GeoFM (**ScaleMAE**). Bold indicates the best mIoU.

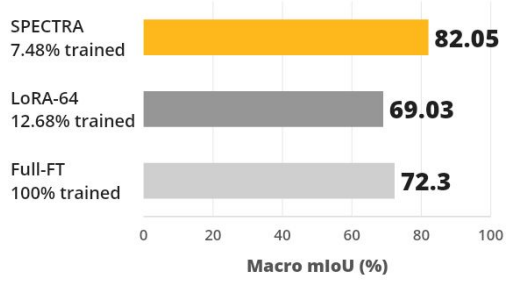
Dataset	Method	Trainable params. %↓	Test macro mIoU (%)↑	Test macro F1 (%)↑	FG IoU (%)↑
GeoFM: ScaleMAE					
Sen1Floods11	LP	4.75%	70.16 ± 4.12	80.35 ± 3.50	48.29 ± 6.80
	LoRA-16	6.73%	76.27 ± 2.21	85.25 ± 1.71	57.92 ± 4.18
	LoRA-32	8.72%	77.07 ± 2.05	85.85 ± 1.58	59.17 ± 3.80
	LoRA-64	12.68%	77.17 ± 1.76	85.93 ± 1.33	59.34 ± 3.18
	Surgical	28.57%	73.39 ± 2.46	82.98 ± 1.99	52.98 ± 4.39
	Full-FT	100.00%	75.41 ± 0.99	84.64 ± 0.80	56.54 ± 1.93
	SPECTRA	7.23%	**85.45 ± 2.08**	**91.74 ± 1.32**	**74.11 ± 3.50**
FireScars	LP	4.75%	67.78 ± 1.12	80.49 ± 0.77	59.31 ± 0.85
	LoRA-16	6.73%	67.86 ± 1.96	80.62 ± 1.38	60.64 ± 1.60
	LoRA-32	8.72%	68.20 ± 1.89	80.86 ± 1.31	61.04 ± 1.29
	LoRA-64	12.68%	69.03 ± 0.91	81.44 ± 0.61	61.58 ± 0.40
	Surgical	28.57%	68.27 ± 1.56	80.90 ± 1.09	60.72 ± 1.31
	Full-FT	100.00%	72.30 ± 1.39	83.72 ± 0.92	65.24 ± 1.17
	SPECTRA	7.48%	**82.05 ± 2.07**	**90.05 ± 1.29**	**77.36 ± 2.94**
Landslide4Sense	LP	4.75%	74.13 ± 0.05	82.93 ± 0.04	50.18 ± 0.11
	LoRA-16	6.73%	75.05 ± 0.20	83.71 ± 0.17	51.89 ± 0.37
	LoRA-32	8.72%	75.16 ± 0.28	83.81 ± 0.23	52.12 ± 0.49
	LoRA-64	12.68%	75.15 ± 0.19	83.79 ± 0.16	52.04 ± 0.33
	Surgical	28.57%	74.82 ± 0.31	83.50 ± 0.26	51.40 ± 0.56
	Full-FT	100.00%	75.36 ± 0.07	83.97 ± 0.06	52.44 ± 0.14
	SPECTRA	6.20%	**76.33 ± 0.06**	**84.77 ± 0.06**	**54.25 ± 0.15**
SA Crop Type	LP	4.75%	30.36 ± 0.42	43.37 ± 0.71	24.44 ± 2.42
	LoRA-16	6.73%	31.50 ± 0.44	44.39 ± 0.60	26.35 ± 1.22
	LoRA-32	8.72%	31.90 ± 0.34	45.08 ± 0.58	25.91 ± 2.16
	LoRA-64	12.68%	32.09 ± 0.35	45.06 ± 0.46	26.74 ± 1.25
	Surgical	28.57%	31.03 ± 0.33	44.02 ± 0.27	25.44 ± 1.58
	Full-FT	100.00%	33.22 ± 0.52	46.58 ± 0.63	29.26 ± 1.84
	SPECTRA	7.56%	**31.97 ± 0.30**	**45.03 ± 0.36**	**29.57 ± 1.55**

Full comparison across all GeoFMs can be found in **Table 4** of our paper.

Sen1Floods11



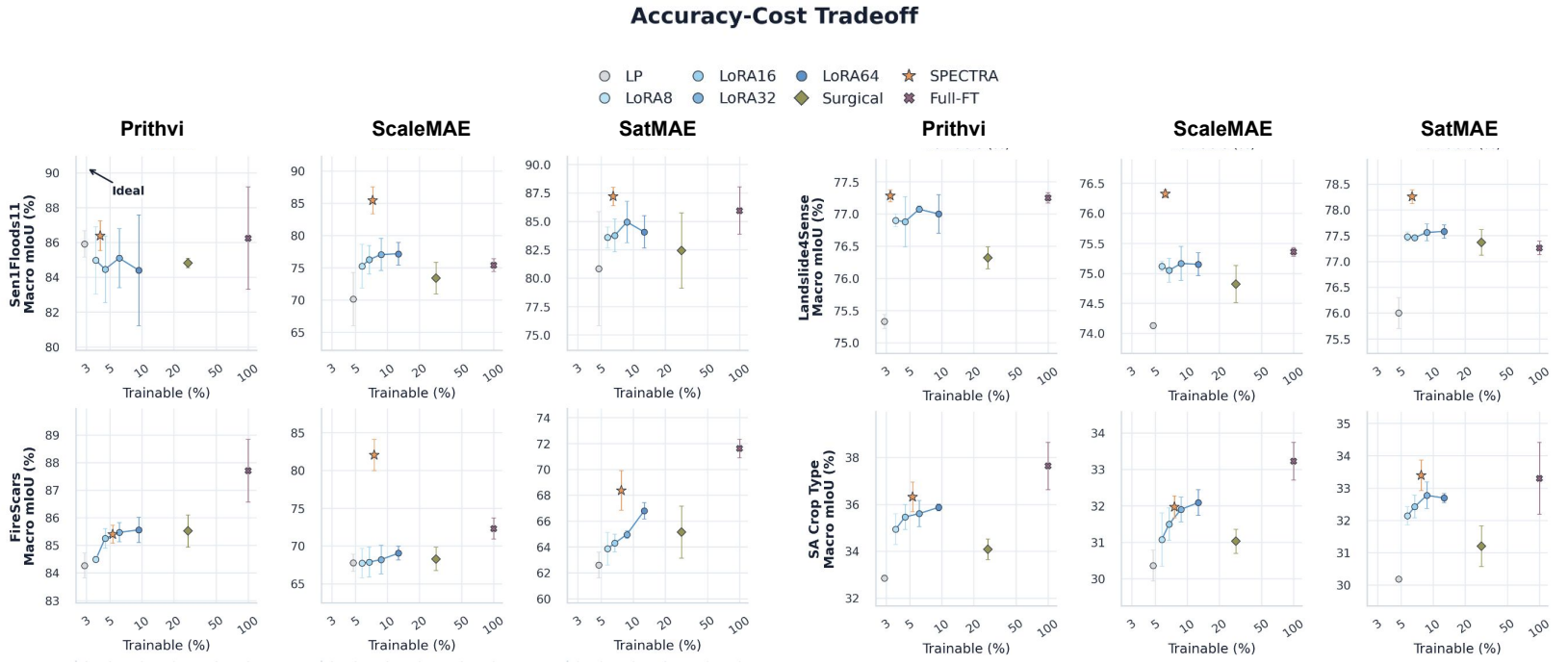
FireScars



Among all GeoFMs, SPECTRA makes the largest improvement on **ScaleMAE. SPECTRA outperforms other fine-tuning pipelines with a small number of parameters to train.**

Results 1: overall comparison across GeoFMs and datasets

We can also see the same conclusion from the accuracy-cost tradeoff plot: **SPECTRA outperforms other fine-tuning pipelines with a relatively small number of trainable model weights.**

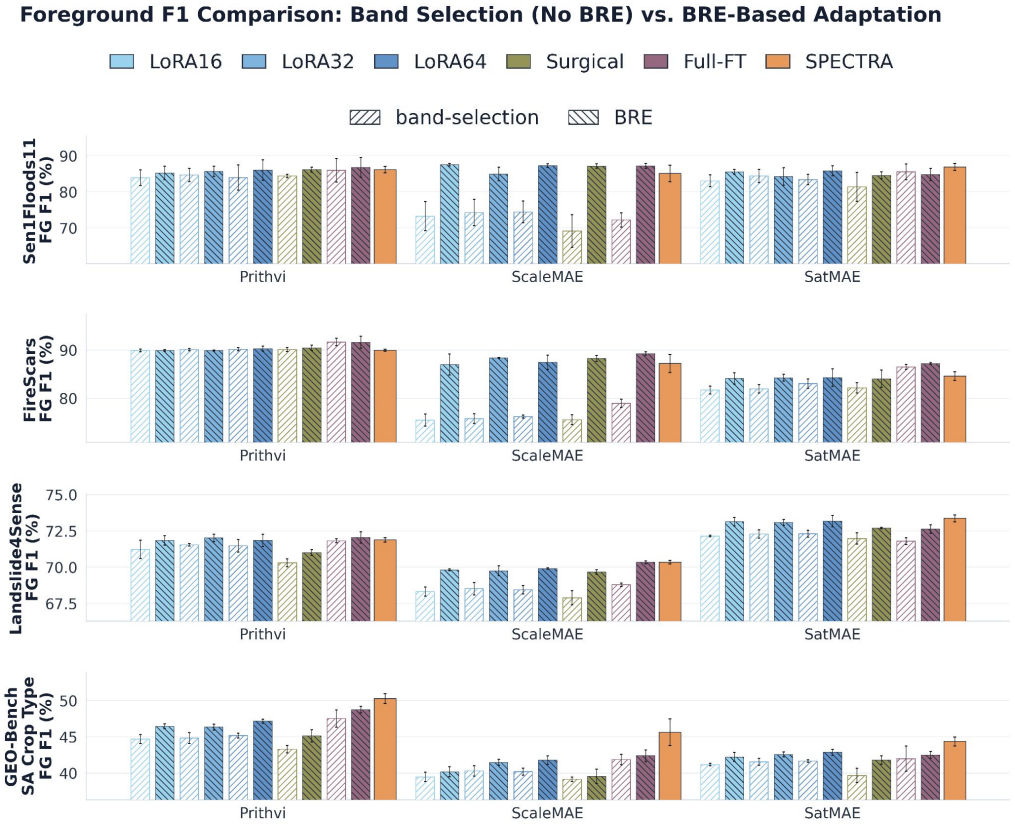


Experiment 2: Ablation Study of BRE Spectral Adaptation

Compare BRE with band selection and MLP projection, when not using ST-LoRA.

Input adapter	Bands used	Mapping to the pretrained input
Band selection	K selected bands	Select bands with the closest central wavelengths
Direct MLP projection	All C bands	Project C channels into K channels
BRE	**All C bands**	**Add an all-band residual to the selected-band anchor**

Results 1: Compare BRE with Band-Selection



- **BRE is the most helpful when band selection discards more task-relevant spectral information.**
  - **Largest gain** in ***ScaleMAE***: pre-trained by **3 RGB** bands only. Band-selection discards all the rest bands.
  - **Smallest gain** in ***Prithvi***: pre-trained by 6 Sentinel-2 bands. FireScars has the same bands; other datasets have full 13 Sentinel-2 bands.

Results 2: Compare BRE with MLP-Projection

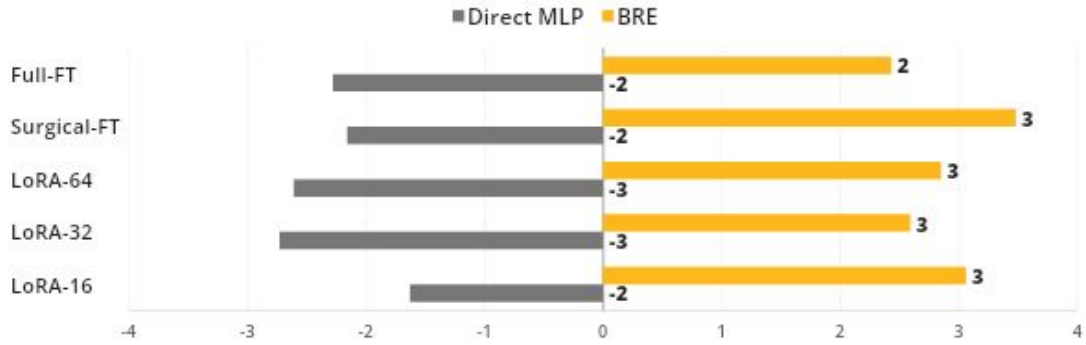
Table. **Comparison between MLP projection and BRE.** Values are average test macro **mIoU (%)** over *n* experiments from 3 GeoFMs and 4 datasets. **Deltas (Δ) are relative to the band-selection.**

Scope	*n*	band-selection	+MLP	+BRE	ΔMLP	ΔBRE
Overall	60	66.36	64.08	**69.24**	−2.28	+2.88
LoRA-16	12	65.89	64.26	**68.95**	−1.63	+3.06
LoRA-32	12	66.32	63.59	**68.90**	−2.73	+2.59
LoRA-64	12	66.45	63.84	**69.30**	−2.61	+2.85
Surgical	12	65.37	63.20	**68.84**	−2.16	+3.48
Full-FT	12	67.77	65.49	**70.20**	−2.28	+2.43

Full comparison can be found in Table 12 - 14 in our paper.

***n*:** The number of experiments averaged.

- 3 GeoFMs x 4 datasets = **12** experiments for each encoder-adaptation method
- 12 x 5 adaptation methods = 60 experiments in total



Overall MLP: −2.28 points

Overall BRE: +2.88 points

Experiment 3: Ablation Study of ST-LoRA

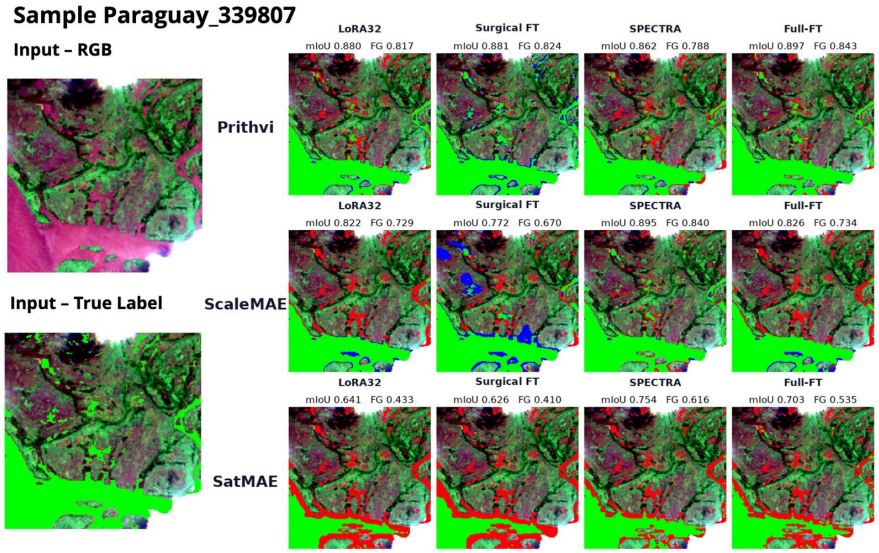
Table 6: Comparison between **automatic ST-LoRA** and **manually selected ranks** on Sen1Floods11

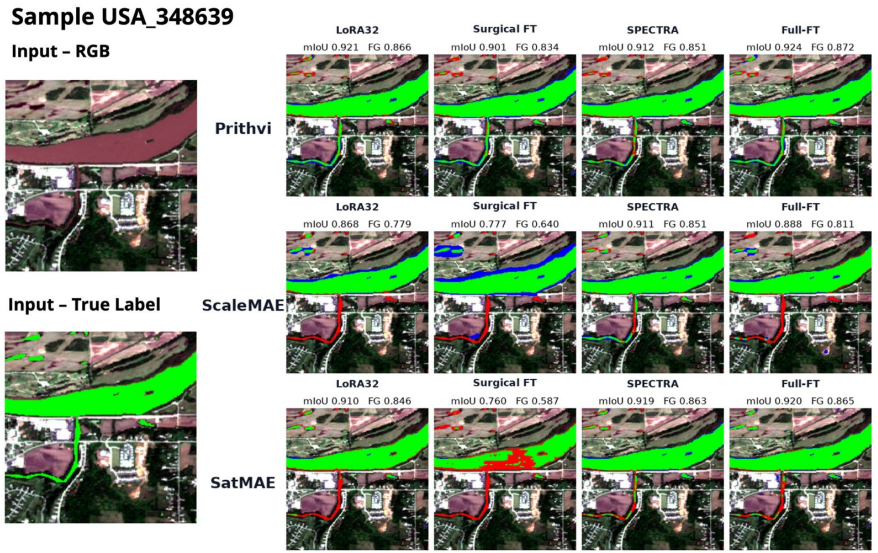
Method & ranks	Rank budget	Trainable params	Macro mIoU %
GeoFM: Prithvi-EO-2.0 600M			
M1 [4,4,8,16]	32	23.94M	86.58 ± 0.71
M2 [16,8,4,4]	32	23.94M	84.98 ± 2.37
M3 [8,16,16,8]	48	26.57M	85.85 ± 2.13
M4 [4,8,16,32]	60	28.53M	85.85 ± 1.47
M5 [32,16,16,8]	72	30.50M	85.02 ± 2.39
M6 [8,16,24,32]	80	31.81M	85.22 ± 1.89
M7 [4,8,16,64]	92	33.77M	86.05 ± 1.05
M8 [64,32,16,8]	120	38.36M	85.15 ± 2.89
ST-LoRA [8,16,16,8]	48	26.35M	85.88 ± 1.24
GeoFM: ScaleMAE			
M1 [4,4,8,16]	32	18.23M	85.27 ± 1.77
M2 [16,8,4,4]	32	18.23M	84.98 ± 1.82
M3 [8,16,16,8]	48	19.80M	84.63 ± 1.75
M4 [4,8,16,32]	60	20.98M	85.15 ± 1.83
M5 [32,16,16,8]	72	22.16M	84.99 ± 1.67
M6 [8,16,24,32]	80	22.95M	85.25 ± 1.68
M7 [4,8,16,64]	92	24.13M	85.12 ± 1.89
M8 [64,32,16,8]	120	26.88M	85.01 ± 1.59
ST-LoRA [16,32,16,16]	80	22.95M	85.29 ± 2.17

- **Our automatic ST-LoRA makes compatible schedules to manually-selected ranks, but is more efficient.**
- **ST-LoRA only need to run once** before fine-tuning. Manually selection might need at most 5^4 runs to select the best combination.

Visualization Examples

SPECTRA recovers foreground regions missed by band selection

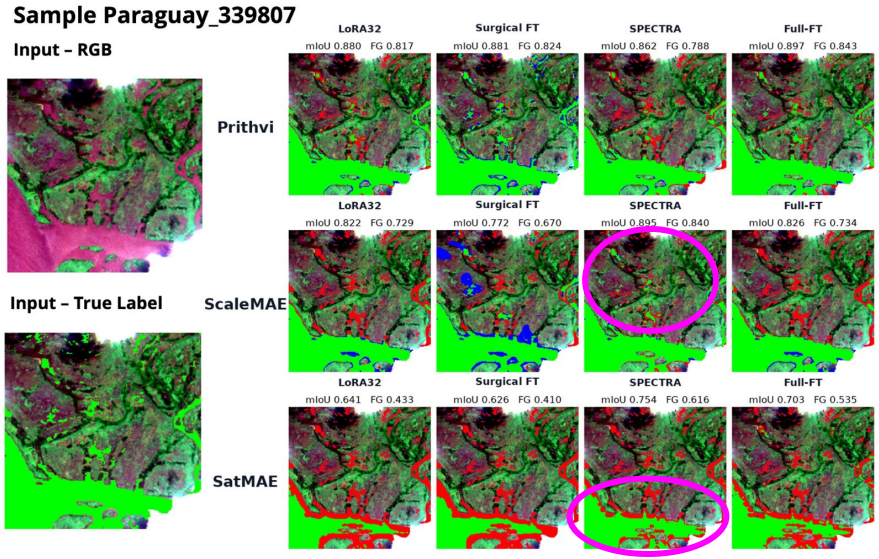


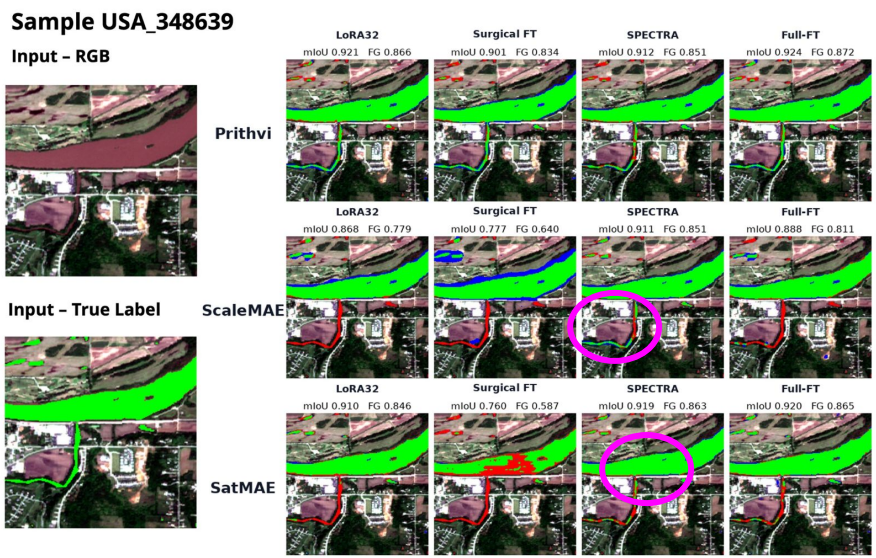


Green: true positive; Blue: false positive; Red: false negative

Visualization Examples

SPECTRA recovers foreground regions missed by band selection





Green: true positive; Blue: false positive; Red: false negative

Takeaways

Cross-sensor transfer needs both spectral and encoder adaptation. We propose BRE and ST-LoRA, respectively.

BRE solves cross-sensor adaptation problem and uses all bands to improve performance.

ST-LoRA reduces model capacity by planning stage-wise LoRA ranks.

+8.28 points

Mean macro mIoU gain across ScalesMAE comparisons

17–34% fewer

Trainable parameters with SPECTRA versus LoRA-32 on Sen1Floods11







arXiv Paper: arXiv:2608.01751

GitHub github.com/big-data-lab-umbc/SPECTRA

UMBC Website and CV: bdal.umbc.edu/people/xingyanli

ScaleMAE: Sen1Floods11

Mean ± standard deviation, retained from the main comparison table

Pipeline	Trainable (%)	Macro mIoU (%)	Macro F1 (%)	FG IoU (%)
LP	4.75	70.16 ± 4.12	80.35 ± 3.50	48.29 ± 6.80
LoRA-16	6.73	76.27 ± 2.21	85.25 ± 1.37	57.92 ± 4.18
LoRA-32	8.72	77.07 ± 2.05	85.85 ± 1.58	59.17 ± 3.80
LoRA-64	12.68	77.17 ± 1.76	85.93 ± 1.33	59.34 ± 3.18
Surgical-FT	28.57	73.39 ± 2.46	82.98 ± 1.99	52.98 ± 4.39
Full-FT	100.00	75.41 ± 0.99	84.64 ± 0.80	56.54 ± 1.93
SPECTRA	**7.23**	**85.45 ± 2.08**	**91.74 ± 1.32**	**74.11 ± 3.50**

ScaleMAE: FireScars

Mean ± standard deviation, retained from the main comparison table

Pipeline	Trainable (%)	Macro mIoU (%)	Macro F1 (%)	FG IoU (%)
LP	4.75	67.78 ± 1.12	80.49 ± 0.77	59.31 ± 0.85
LoRA-16	6.73	67.86 ± 1.96	80.62 ± 1.38	60.64 ± 1.60
LoRA-32	8.72	68.20 ± 1.89	80.86 ± 1.31	61.04 ± 1.29
LoRA-64	12.68	69.03 ± 0.91	81.44 ± 0.61	61.58 ± 0.40
Surgical-FT	28.57	68.27 ± 1.56	80.90 ± 1.09	60.72 ± 1.31
Full-FT	100.00	72.30 ± 1.39	83.72 ± 0.92	65.24 ± 1.17
SPECTRA	**7.48**	**82.05 ± 2.07**	**90.05 ± 1.29**	**77.36 ± 2.94**

ScaleMAE: Landslide4Sense

Mean ± standard deviation, retained from the main comparison table

Pipeline	Trainable (%)	Macro mIoU (%)	Macro F1 (%)	FG IoU (%)
LP	4.75	74.13 ± 0.05	82.93 ± 0.04	50.18 ± 0.11
LoRA-16	6.73	75.05 ± 0.20	83.71 ± 0.17	51.89 ± 0.37
LoRA-32	8.72	75.16 ± 0.28	83.81 ± 0.23	52.12 ± 0.49
LoRA-64	12.68	75.15 ± 0.19	83.79 ± 0.16	52.04 ± 0.33
Surgical-FT	28.57	74.82 ± 0.31	83.50 ± 0.26	51.40 ± 0.56
Full-FT	100.00	75.36 ± 0.07	83.97 ± 0.06	52.44 ± 0.14
SPECTRA	**6.20**	**76.33 ± 0.06**	**84.77 ± 0.06**	**54.25 ± 0.15**

ScaleMAE: SA Crop Type

Mean ± standard deviation, retained from the main comparison table

Pipeline	Trainable (%)	Macro mIoU (%)	Macro F1 (%)	FG IoU (%)
LP	4.75	30.36 ± 0.42	43.37 ± 0.71	24.44 ± 2.42
LoRA-16	6.73	31.50 ± 0.44	44.39 ± 0.60	26.35 ± 1.22
LoRA-32	8.72	31.90 ± 0.34	45.08 ± 0.58	25.91 ± 2.16
LoRA-64	12.68	32.09 ± 0.35	45.06 ± 0.46	26.74 ± 1.25
Surgical-FT	28.57	31.03 ± 0.33	44.02 ± 0.27	25.44 ± 1.58
Full-FT	100.00	33.22 ± 0.52	46.58 ± 0.63	29.26 ± 1.84
SPECTRA	**7.56**	**31.97 ± 0.30**	**45.03 ± 0.36**	**29.57 ± 1.55**

Prithvi: manual and automatic rank schedules

Sen1Floods11 comparison from Table 6

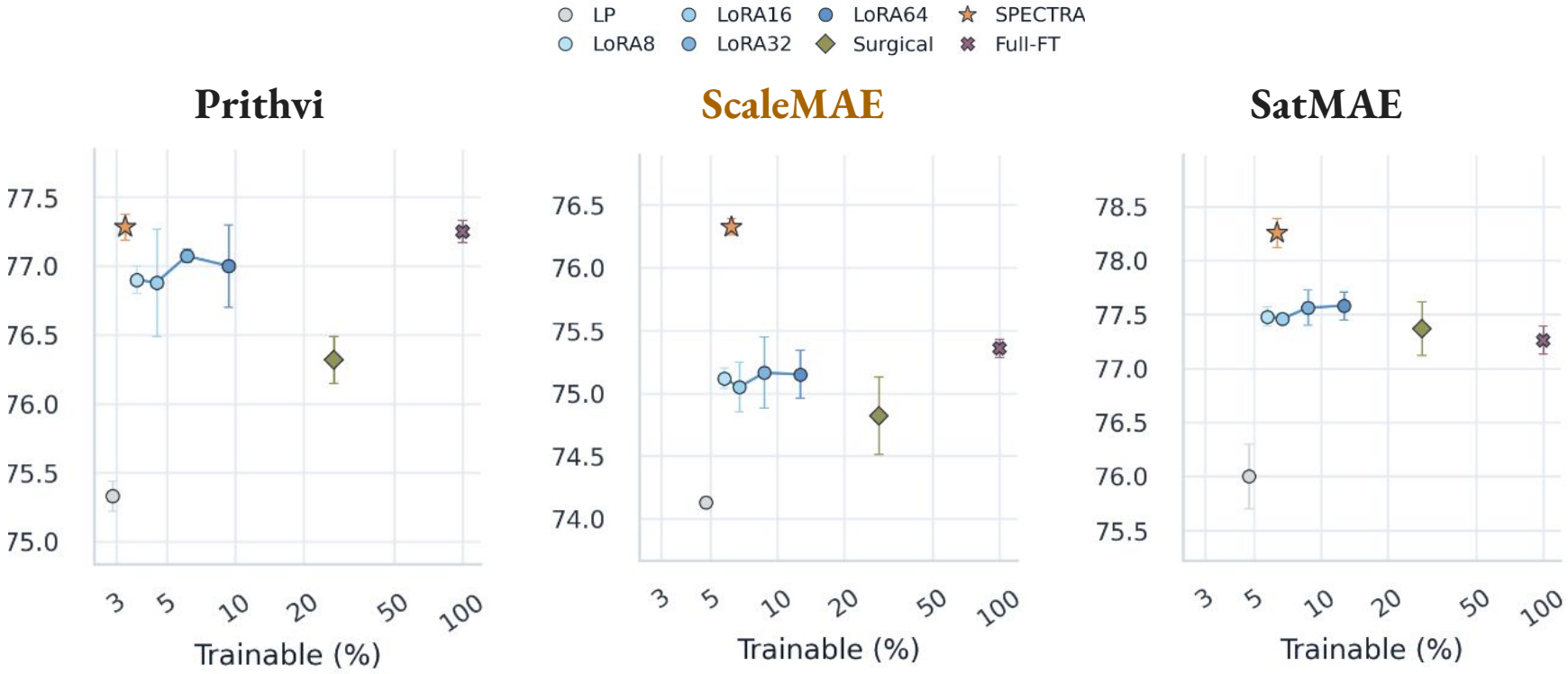
Method	Stage ranks	Rank budget	Trainable (M)	Macro mIoU (%)
M1	[4,4,8,16]	32	23.94	86.58 ± 0.71
M2	[16,8,4,4]	32	23.94	84.98 ± 2.37
M3	[8,16,16,8]	48	26.57	85.85 ± 2.13
M4	[4,8,16,32]	60	28.53	85.85 ± 1.47
M5	[32,16,16,8]	72	30.50	85.02 ± 2.39
M6	[8,16,24,32]	80	31.81	85.22 ± 1.89
M7	[4,8,16,64]	92	33.77	86.05 ± 1.05
M8	[64,32,16,8]	120	38.36	85.15 ± 2.89
ST-LoRA	**[8,16,16,8]**	**48**	**26.35**	**85.88 ± 1.24**

ScaleMAE: manual and automatic rank schedules

Sen1Floods11 comparison from Table 6

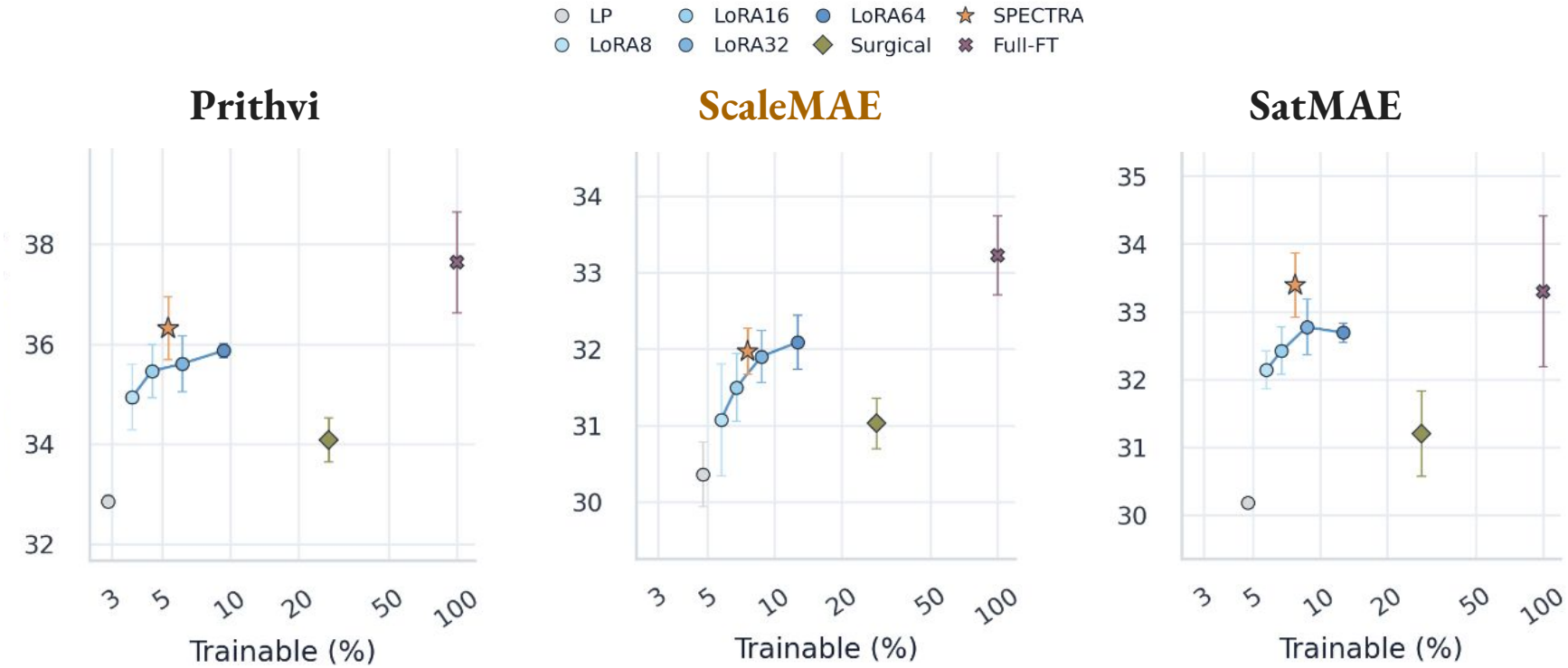
Method	Stage ranks	Rank budget	Trainable (M)	Macro mIoU (%)
M1	[4,4,8,16]	32	18.23	85.27 ± 1.77
M2	[16,8,4,4]	32	18.23	84.98 ± 1.82
M3	[8,16,16,8]	48	19.80	84.63 ± 1.75
M4	[4,8,16,32]	60	20.98	85.15 ± 1.83
M5	[32,16,16,8]	72	22.16	84.99 ± 1.67
M6	[8,16,24,32]	80	22.95	85.25 ± 1.68
M7	[4,8,16,64]	92	24.13	85.12 ± 1.89
M8	[64,32,16,8]	120	26.88	85.01 ± 1.59
ST-LoRA	**[16,32,16,16]**	**80**	**22.95**	**85.29 ± 2.17**

Accuracy and parameter cost: Landslide4Sense



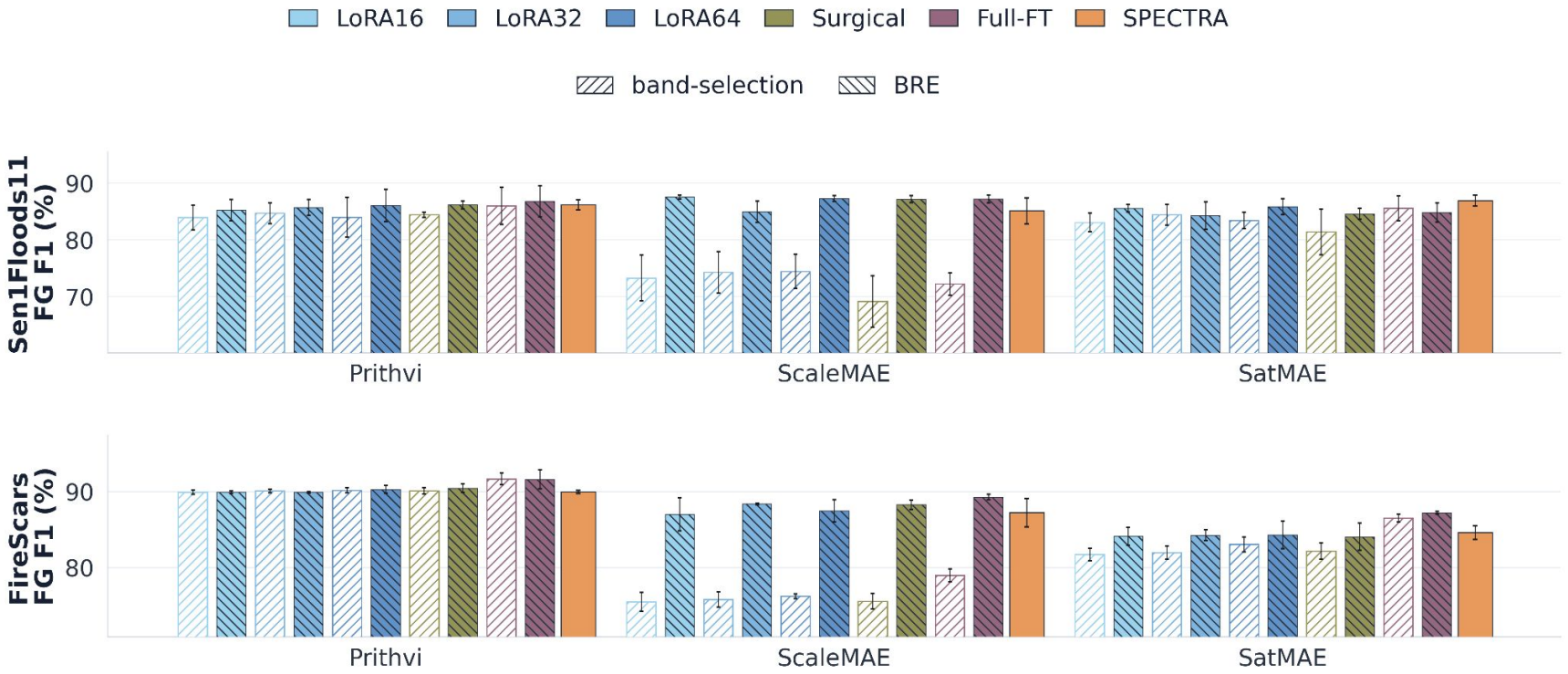
Vertical: macro mIoU (%) Horizontal: trainable parameters (%) on a log scale

Accuracy and parameter cost: SA Crop Type

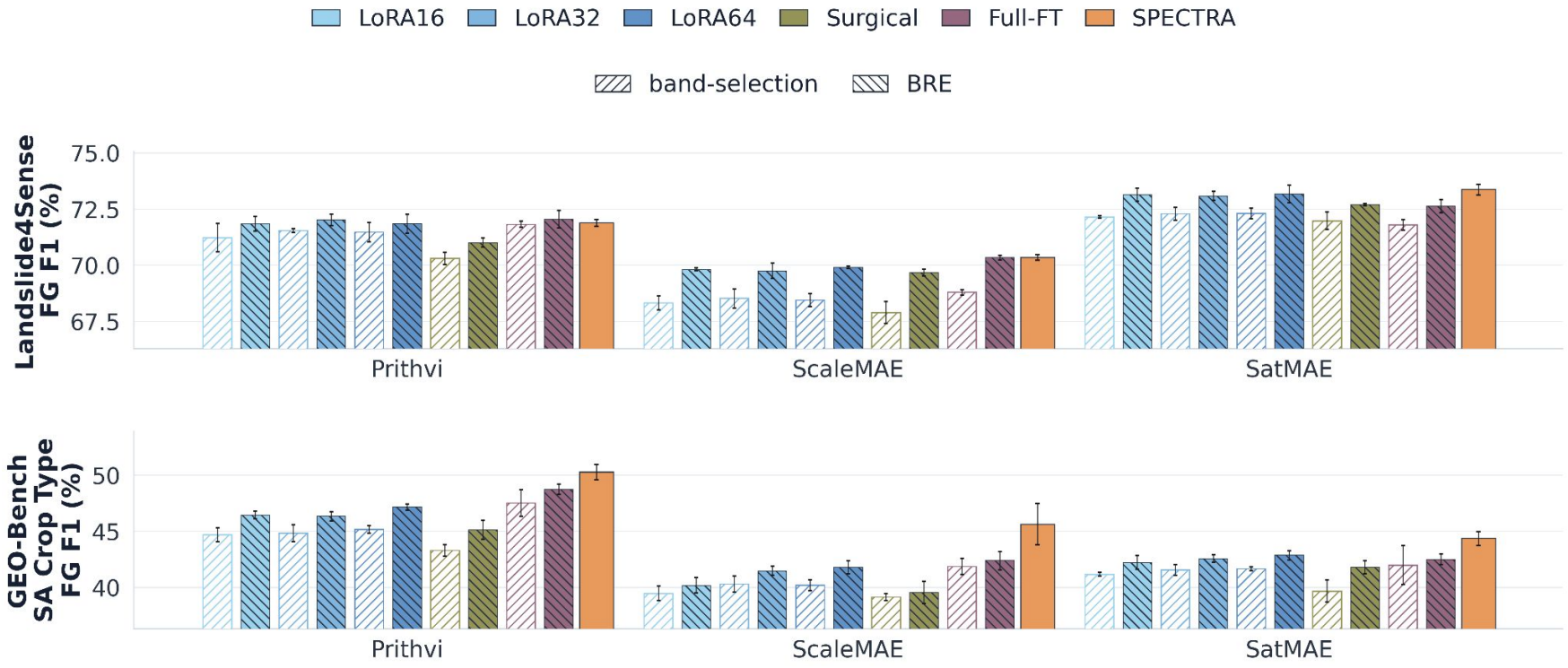


Vertical: macro mIoU (%) Horizontal: trainable parameters (%) on a log scale

Foreground F1: Floods and fire scars



Foreground F1: Landslides and crop type



BRE and direct MLP: complete ablation table

Mean test macro mIoU (%). Deltas are relative to band selection.

Scope	n	Band selection	Direct MLP	BRE	Δ MLP	Δ BRE
Overall	**60**	**66.36**	**64.08**	**69.24**	**−2.28**	**+2.88**
LoRA-16	12	65.89	64.26	68.95	−1.63	+3.06
LoRA-32	12	66.32	63.59	68.90	−2.73	+2.59
LoRA-64	12	66.45	63.84	69.30	−2.61	+2.85
Surgical-FT	12	65.37	63.20	68.84	−2.16	+3.48
Full-FT	12	67.77	65.49	70.20	−2.28	+2.43

Sen1Floods11: parameter and accuracy comparison

Uniform LoRA-32 versus the complete SPECTRA pipeline

GeoFM	LoRA-32 trainable	SPECTRA trainable	Relative reduction	LoRA-32 mIoU	SPECTRA mIoU
Prithvi	6.12%	4.06%	33.7%	85.10	86.39
ScaleMAE	8.72%	7.23%	17.1%	77.07	85.45
SatMAE	8.72%	6.49%	25.6%	84.95	87.19

Accuracy changes reflect BRE and ST-LoRA together.